Label Reliability and Study-Level Leakage Bound Measurable Performance in T-Cell Epitope Immunogenicity Prediction

I Nyoman Agung Ananda Terano[1] **Mirrabell Frederica Hadiwijono**[2]

Abstract

Predictors of CD8⁺ T-cell epitope immunogenicity report ROC-AUC values of 0.85 and above under random cross-validation on Immune Epitope Database (IEDB) data, and are used to prioritise vaccine candidates. We ask what those labels can support, and how benchmark construction shapes the result. Among 65,732 human MHC-I T-cell assay records, 2,914 peptide–HLA pairs were assayed independently by two or more publications; one laboratory's outcome predicts another's with 72.2% agreement, κ of 0.273, and an AUC-equivalent of 0.636 (95% CI 0.617–0.657), or 0.681 for pairs assayed by exactly two studies. Because a repeat experiment observes the biology directly, reported performance exceeds what these labels can measure. Peptide splitting, the field's standard precaution, lowers AUC by only 0.014 (0.826 against 0.840 for random splitting), whereas holding out source publications gives 0.607; grouping near-identical or overlapping peptides changes it by under 0.02, so sequence similarity is not the cause. One all-negative study supplies 60% of that gap. An allele-identity baseline that never sees the peptide reaches 0.752, above the 0.606, 0.647 and 0.559 of three published predictors on the same pairs. DeepImmuno and BigMHC_IM had been trained on 37.5% and 27.1% of those pairs and fall by 0.23 and 0.26 AUC on pairs they had not seen. Counting untested candidates as negatives rather than assayed negatives raises per-patient AUC from ≈0.55 to ≈0.97. Under leakage-free evaluation our best model reaches 0.573 (0.544–0.603), 84% of the conservative bound, regardless of architecture. We release the benchmark, both split schemes, all predictions, and a reporting protocol.

Keywords: T-cell epitope · immunogenicity prediction · data leakage · benchmark design · label reliability · reproducibility · IEDB

[1]Institut Teknologi Sepuluh Nopember, Surabaya, Indonesia. [2]University of Toronto, Toronto, Canada. Correspondence to: I Nyoman Agung Ananda Terano (teranoace@gmail.com).

1. Introduction

Predictors of which peptide–MHC complexes trigger a CD8⁺ T-cell response are used to design neoantigen vaccines and to choose targets for adoptive cell therapy. Personalised neoantigen vaccines have reached randomised trials in melanoma and pancreatic cancer (Ott et al., 2017; Sahin et al., 2017; Rojas et al., 2023; Weber et al., 2024), and which candidates are synthesised rests on predictions of this kind. Experimental screening is slow and costly, so predictors trained on IEDB have multiplied, and recent ones report ROC-AUC values of 0.85 and above under random cross-validation (**Table 1**). Neoantigen pipelines lacked reference data for comparison until a consortium experimentally tested 608 predicted epitopes (Wells et al., 2020). Two established facts argue against taking those numbers at face value.

Reported performance tracks the evaluation protocol more closely than it tracks the method (**Table 1**). Under random cross-validation on pooled data, reported figures reach 0.85 to 0.94. Where an independently constructed test set is used, the same class of model scores between 0.56 and 0.68. The first sequence-based model of immunogenicity reported 0.65, and 0.61 once immunogenic and non-immunogenic peptides were matched for binding affinity (Calis et al., 2013). Lee et al. (2023) attribute the higher recent figures to dataset composition, noting that "high reported performance from the latest models incorporating peptide-HLA pairs might be driven by HLA bias from an imbalanced dataset". The reported numbers therefore do not form a progression: they separate by how the data were split. This paper measures why, and what the labels themselves permit.

First, peptide and allele do not determine the outcome. A response also depends on the individual's T-cell receptor repertoire, prior antigen exposure, the rest of their HLA genotype and the tissue context (Lu et al., 2026). Two people with the same restricting allele can respond differently to the same peptide, so a model given only a peptide and an allele has a ceiling below 1.0 whatever its architecture.

Second, laboratories measure the response inconsistently. International proficiency panels run by the Cancer Vaccine Consortium found that laboratories differ substantially on the same material. In two ELISpot panels (36 laboratories in 2005, 29 in 2006) performance was scored by the ability to detect all predefined responses correctly, and it improved significantly in the second panel once protocol lessons were applied (Janetzki et al., 2008). A 27-laboratory HLA-peptide multimer panel found

Table 1. Reported performance of sequence-based CD8⁺ T-cell immunogenicity predictors, grouped by the evaluation protocol that produced it. All values are ROC-AUC as reported by the cited work; the verbatim source sentence for each is recorded in *data/external/reported_performance.csv*. Our own estimates are shown for comparison.

Predictor	Reported	Evaluation protocol
Random cross-validation on pooled data		
TRAP, self-antigen model (Lee et al., 2023)	0.943	10-fold cross-validation
DeepImmuno (Li et al., 2021)	0.85	10-fold cross-validation on its IEDB training data
TRAP, pathogenic model (Lee et al., 2023)	0.764	10-fold cross-validation
Calis et al. (2013)	0.65	cross-validation; 0.61 with binding affinity matched
Held-out or independently constructed test set		
NeoaPred (Jiang et al., 2024)	0.81	held-out test set; adds pHLA surface and structural features
INeo-Epp (Wang et al., 2020)	0.77	external validation set (0.81 at its internal optimum)
T-SCAPE-IM (Kim et al., 2025)	0.678	newly constructed test set
Modular protein language model (O'Brien et al., 2024)	0.619	pMHC immunogenicity benchmark
BigMHC_IM, scored by Kim et al. (2025)	0.569	independent evaluation
MHCnuggets 2.4, scored by Kim et al. (2025)	0.557	independent evaluation
This study		
Measured reliability of the labels (§3.2)	0.636–0.681	repeat experiments on the same pair
Leakage-free estimate (§3.8)	0.573	within study and allele, study-grouped

significant differences across laboratories in both assay performance and the frequencies of specific T cells reported (Britten et al., 2009).

Harmonisation guidelines aim to reduce this variation in prospective, coordinated studies (Janetzki et al., 2008; Britten et al., 2009), but they cannot be applied retroactively to a database that aggregates decades of heterogeneous protocols. To our knowledge, what that heterogeneity does to machine-learning benchmarks built on IEDB has not been measured. Predictors deal with label disagreement by convention. PredIG, for example, calls a peptide–HLA pair positive if any experiment found a responder and discards the conflicting negative instances (Farriol-Duran et al., 2025); this is the any-positive rule whose effect on reliability we measure in §3.3.

Related problems have been reported. Lee et al. documented HLA bias in immunogenicity predictors and showed that cross-species splits lower performance (Lee et al., 2023). A benchmark of 50 TCR–epitope predictors found that all of them struggle to generalize to unseen epitopes, and cautioned that adding external TCRs as negatives would "introduce batch-like confounders, causing models to learn dataset-specific artifacts rather than true TCR–epitope binding signals" (Lu et al., 2026). Tools also treat the separation of training from test data inconsistently: T-SCAPE excludes fragments with fewer than two mismatches, and notes that NetMHCpan removes only exact peptide matches and that BigMHC allows overlapping peptide sequences when the allele differs (Kim et al., 2025). Kapoor and Narayanan reviewed 17 scientific fields and found leakage affecting 294 papers; in one case study, complex models claimed to beat logistic regression stopped doing so once leakage was corrected (Kapoor and Narayanan, 2023).

We are not aware of a published measurement of what the labels themselves can support, or of which construction choices open the gap between that limit and reported performance. This paper does both. Our contributions are:

1. A measured reliability bound for IEDB immunogenicity labels, with sensitivity analyses over aggregation rule, retesting frequency, assay family and HLA locus.
2. A demonstration that peptide-level splitting does little to prevent leakage, with controls for sequence similarity, model class, assay family and the single largest study, replicated in direction on an independent dataset, and identification of source publication as the grouping unit that determines the result.
3. A contamination audit of two widely used published predictors against our evaluation set.

4. A leakage-free performance estimate, an architecture comparison showing the choice makes no measurable difference, and a worked example of a feature that standard validation endorses and grouped validation rejects.

2. Scope and definitions

Scope. We examine benchmarks for $CD8^+$ T-cell epitope immunogenicity built from IEDB assay records, where a model receives a peptide and an MHC-I allele and predicts whether a T-cell response was observed. We do not examine MHC binding or ligand presentation prediction, whose labels come from binding or mass-spectrometry measurements rather than from a T-cell assay, and where pan-allele predictors such as NetMHCpan-4.1 and MHCflurry 2.0 report far higher figures on their own task (Reynisson et al., 2020; O'Donnell et al., 2020). We also leave out patient-specific neoantigen prioritisation pipelines, which use tumour sequencing and RNA expression in addition to peptide sequence (Müller et al., 2023); §3.7 shows why conflating the two inflates apparent performance.

Grouping unit. A split is defined by the unit held out. Splitting *by peptide* means no peptide appears in both training and test data; splitting *by study* means no record from a given publication appears in both. Leakage here is shared structure: records from one publication share a cohort, a protocol and often an outcome, so a model can identify the source rather than the biology. We use "grouping unit" throughout to name the choice, because it is the choice that determines the reported number (§3.4).

Label reliability. We define the reliability of a label set as the agreement between two independent measurements of the same quantity, here two publications assaying the same peptide–HLA pair. Expressed as an AUC, this is the score a repeat experiment achieves when used to predict another experiment's outcome. Because a repeat experiment observes the biology rather than inferring it from sequence, it is the most informative predictor obtainable for that pair.

Limits of the bound. The reliability figure limits what can be *measured* against a given label set. It does not limit how accurate immunogenicity prediction could be in principle: when annotators carry independent noise, a well-specified model can exceed their mutual agreement (Richie et al., 2022). We return to this in §4.

Operational task. We take the task that matters for vaccine design to be ranking candidates *within one experiment*: given the peptides a laboratory is considering for one patient or one antigen, which should be screened first. This motivates reporting within-study and within-allele AUC alongside pooled figures (§7.5).

3. Results

3.1 One study supplies a third of the data, and most studies report a single outcome

After retrieval and cleaning (§7.1), the benchmark comprises 31,274 unique peptide–HLA pairs, 31.7% positive, drawn from 1,365 publications. Its composition is uneven (**Figure 1**). A single study contributes 11,486 pairs, 37% of the dataset, and every one of its outcomes is negative; the ten largest studies supply 54% of all pairs and the hundred largest 81%. Across all studies, 75% report only one outcome class, and those studies account for 57% of the pairs. Study identity is therefore highly informative about the label before any peptide is examined, which motivates the analyses that follow.

3.2 Immunogenicity labels reproduce poorly between laboratories

Restricting to the 2,914 pairs assayed by two or more distinct publications, and taking each study's majority outcome, we asked how well one study's result predicts another's on the same pair (§7.3, **Table 2**).

Table 2. Agreement between independent publications assaying the same peptide–HLA pair (n = 2,914). The AUC is the score one study's outcome achieves when used to predict another's; because both labels are binary it equals balanced accuracy. Intervals bootstrap over pairs (§7.3).

Quantity	Value (95% CI)
Agreement	0.722 (0.706–0.738)
Cohen's κ	**0.273** (0.235–0.312)
AUC	**0.636** (0.617–0.657)

A κ of 0.273 (Cohen, 1960) denotes fair agreement only on the conventional scale (Landis and Koch, 1977). The agreement between two experiments on the same pair bounds what can be *measured* against these labels.

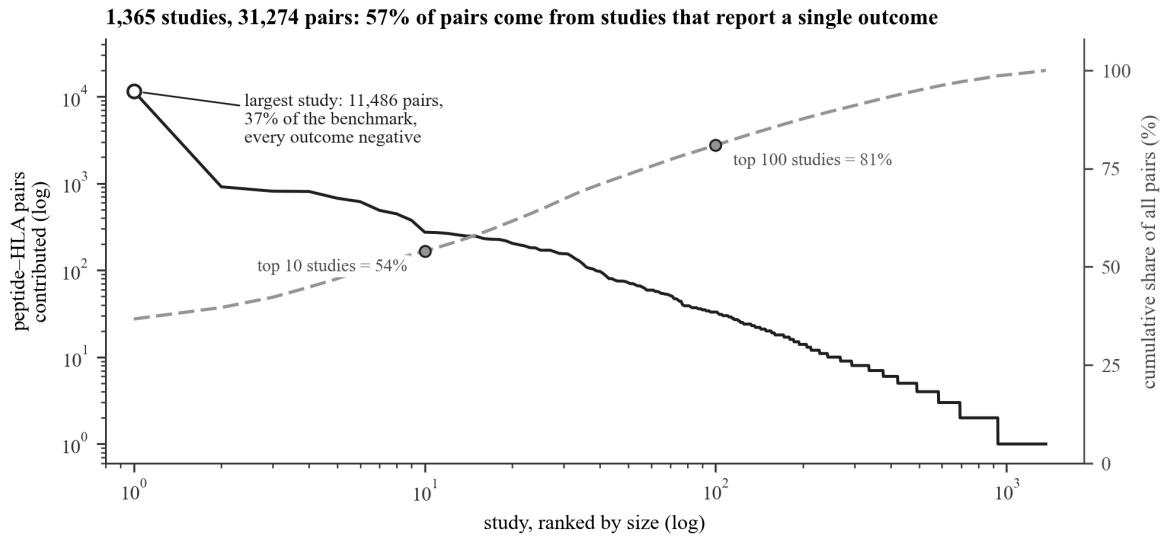


Figure 1. How the benchmark is distributed across its source studies. Pairs contributed per study against study rank, both axes logarithmic (solid, left axis), with the cumulative share of all pairs (dashed, right axis). The largest study contributes 11,486 pairs, 37% of the benchmark, and reports no positive outcome; the ten largest supply 54% of the data and the hundred largest 81%. Studies reporting a single outcome class account for 57% of all pairs.

3.3 The reliability estimate is robust to how it is measured

We tested three threats to this estimate (**Figure 2**, panel a).

Aggregation rule. Majority vote gives 0.636; any-positive 0.652; ties-to-positive 0.646; ties-to-negative 0.556; and raw records with no aggregation at all 0.595. The range is 0.556–0.652, and the least-processed variant falls below the headline figure.

Selection bias. Pairs are not retested at random. Reproducibility declines with retesting frequency: 0.681 for pairs assayed by exactly two studies (n = 1,806), 0.530 for exactly three, 0.543 for four or more. We therefore adopt 0.681 as a conservative bound. Separately, repeat-tested pairs are 72.5% positive against 26.5% for singly-tested pairs, so the decision to retest is driven by the previous result.

Slice dependence. By assay family, ELISpot 0.702, ICS 0.655, multimer/tetramer 0.605; by locus, HLA-A 0.594, HLA-B 0.775. Every slice falls far below reported performance. Multimer and tetramer assays are the least reproducible family here, which fits the significant between-laboratory differences found in a multimer proficiency panel (Britten et al., 2009), although the two measurements are not directly comparable.

3.4 Peptide-level splitting does not prevent leakage

Holding out data by peptide is the standard precaution in this field. We compared grouping units using one model class throughout, so that only the split changes (**Table 3**).

Table 3. Pooled AUC by grouping unit, on IEDB and on an independent dataset (TRAP). One model class is used throughout, so only the held-out unit changes. TRAP records antigen and organism rather than source publication, so the organism is its widest available unit.

Grouping unit	IEDB (this study)	TRAP (independent) (Lee et al., 2023)
Random	0.840	0.857
Peptide	**0.826**	**0.862**
Antigen	—	0.817
Source (study / organism)	**0.607**	**0.552**

On the independent dataset, peptide grouping scores *above* random splitting. On IEDB it lowers AUC by only 0.014 (95% CI 0.009–0.024, resampling studies). In both datasets, widening the grouping unit to the data source collapses performance. Peptide grouping leaves most of the structure that inflates these results untouched.

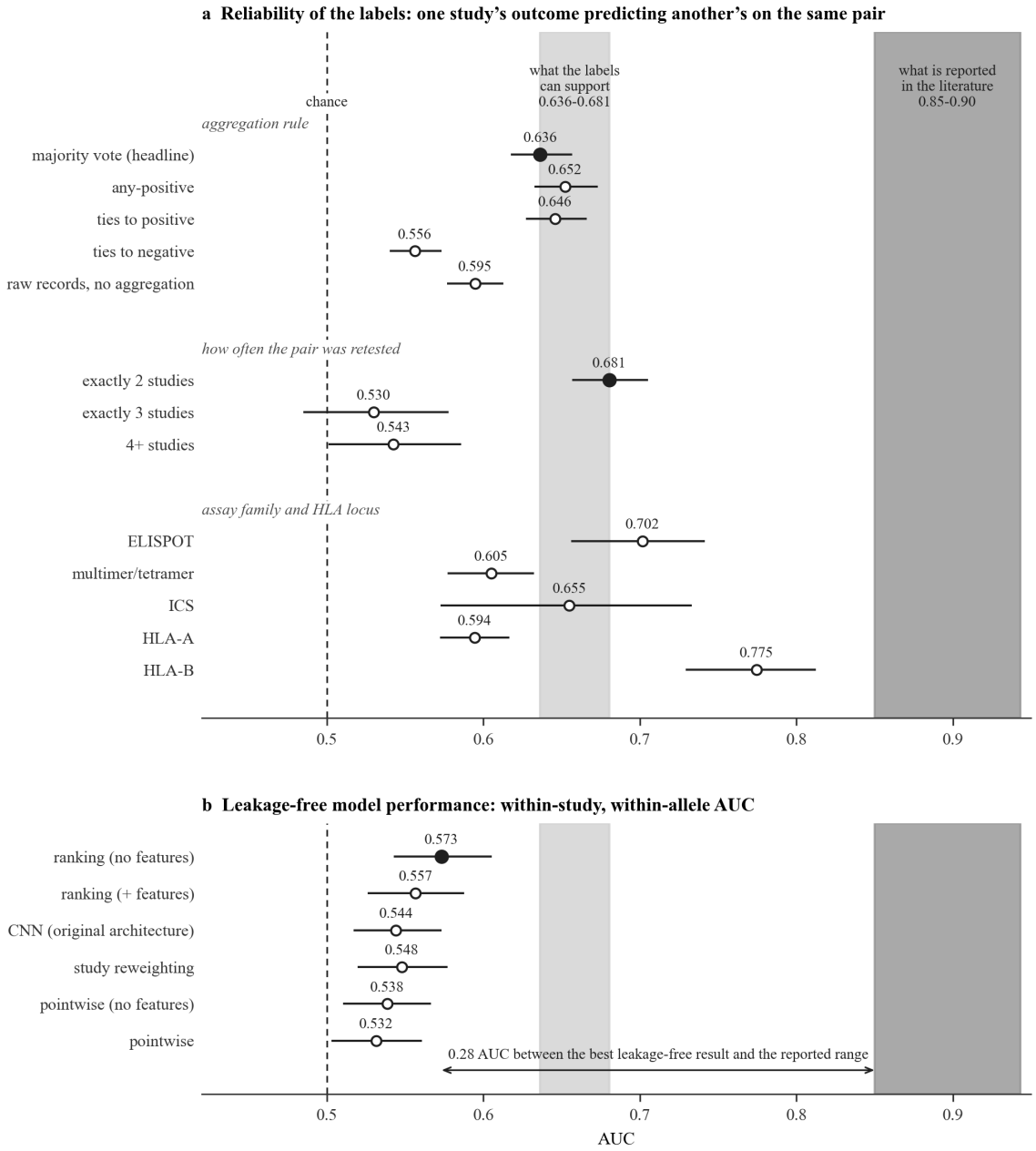


Figure 2. What the labels can support, and what a leakage-free model achieves. Both panels are AUC on one scale, so the two reference bands run through both and are labelled once: the light band is the measured reliability of the labels (0.636– 0.681), the dark band the 0.85–0.94 range reported under random cross-validation (**Table 1**). The dashed line is chance. **(a)** AUC of one study's outcome predicting another's on the same pair, with 95% confidence intervals, under three robustness checks: the rule used to aggregate repeat records, how often the pair was retested, and the assay family and HLA locus. Filled markers are the headline estimate (0.636) and the conservative bound adopted in the text (0.681), which are the edges of the light band. No variant approaches the dark one. **(b)** Within-study, within-allele AUC for six training schemes, with intervals bootstrapped over studies. Every interval clears chance; the best reaches 84% of the conservative bound from panel (a) and leaves 0.28 AUC between itself and the reported range.

Four further checks ask whether this is an artefact of how "peptide" was defined, of the model, of one study, or of mixing assays. Each cell of **Table 4** is pooled AUC from five-fold grouped cross-validation on IEDB unless stated.

Sequence similarity is not the mechanism. Holding out peptides within one or two mismatches of each other, or sharing a six-residue stretch (which groups overlapping windows of one source protein), lowers AUC by 0.008 to 0.010 more than exact-peptide grouping (six-residue rule: 0.008, 95% CI 0.005–0.017). Holding out source publications lowers it by a further 0.210. The model is not recognising near-copies of training peptides; it is recognising the study.

Table 4. Pooled AUC under similarity-based grouping rules and across model classes. The allele prior uses no peptide information. Sequence-similarity rules move AUC by under 0.02 relative to exact-peptide grouping, whereas holding out source publications moves it by more than 0.2.

Grouping unit	Gradient boosting	Within-allele	Logistic regression	Allele prior (no peptide)
Random	0.840	0.768	0.742	0.755
Exact peptide	**0.826**	**0.740**	**0.739**	**0.756**
Peptides within 1 mismatch	0.816	0.718	0.736	0.754
Peptides within 2 mismatches	0.818	0.723	0.736	0.753
Peptides sharing a 6-residue stretch	0.818	0.724	0.735	0.754
Source publication	**0.607**	**0.435**	**0.548**	**0.465**

It is not specific to gradient boosting. Logistic regression gives the same ordering (0.739 under peptide grouping, 0.548 under source grouping). The allele prior, which sees no peptide, reaches 0.756 under peptide grouping and 0.465 under source grouping. Allele frequency predicts the outcome only because studies differ in which alleles they test and what they find; when studies are held out, the same signal points the wrong way.

One study accounts for most of the gap, but not all of it. The single all-negative study (§3.1) inflates the random and peptide-grouped scores. Without it (19,788 pairs remain), random and peptide grouping score 0.765 and 0.759 and source grouping scores 0.673. The random-to-source gap falls from 0.232 (95% CI 0.082–0.296) to 0.092 (0.066–0.118), and the peptide-to-source gap from 0.218 (0.066–0.283) to 0.085 (0.061–0.111). About 60% of the pooled gap therefore comes from this one study. The remainder is smaller but its interval excludes zero, so the effect does not rest on one study alone. It does mean the size of the gap is a property of this benchmark's composition and should not be carried over to others.

Mixing assay types does not explain it. Among pairs whose records all come from one assay family, source grouping still collapses performance, from 0.875 to 0.652 for ELISpot (16,888 pairs, 285 studies), from 0.810 to 0.649 for multimer and tetramer assays (6,273 pairs, 174 studies), and from 0.726 to 0.452 for intracellular cytokine staining (1,343 pairs, 100 studies, so less precise).

A forward-in-time test agrees but is imprecise. Studies were ordered by PubMed ID as a proxy for publication date. Models trained on the earlier 80% of dated pairs (14,484 pairs, PubMed ID up to 36,211,805) and were tested on the 3,611 pairs of 218 later studies, of which 65% are positive. Gradient boosting reaches 0.563 (95% CI 0.442–0.684, resampling studies), logistic regression 0.536 (0.412–0.671) and the allele prior 0.457 (0.381–0.555). These sit with the study-grouped estimates and the gradient boosting interval excludes the 0.82–0.84 of random and peptide splits, but the intervals also include chance, because 50% of the test pairs come from ten studies (the largest holds 274 pairs). Only 18,095 of 31,274 pairs carry a PubMed ID, so the check covers 58% of the benchmark.

3.5 A baseline that cannot see the peptide outscores published predictors

On the held-out test split (4,641 pairs, 1,470 positive), an allele-prior baseline reaches 0.752. It predicts each allele's training-set positive rate and uses no peptide information whatsoever. On the same pairs BigMHC_IM reaches 0.647, DeepImmuno 0.606 and MixMHCpred 0.559: every published predictor scores below the baseline that never sees the peptide (**Table 5**).

The two DeepImmuno rows are different models and the gap between them is itself a result: the published checkpoint scores 0.606, while the same architecture retrained on our training split reaches 0.792. The architecture is not what separates them; exposure to this dataset's study structure is.

The allele prior is constant within an allele, so its within-allele AUC is 0.500 by construction: the entire 0.752 is allele frequency. Holding the allele fixed costs the models trained here between 0.06 and 0.13 AUC, which is the allele signal they had been using. It costs the externally trained predictors nothing, and DeepImmuno and MixMHCpred score marginally *higher* within allele than pooled, because they never learned this dataset's allele skew in the first place. What looks like an advantage for the local models over published ones is, to a large extent, that signal.

A peptide-blind baseline that beats published predictors is diagnostic: on this benchmark, allele identity and dataset structure carry more signal than the peptide sequence does.

Table 5. Performance on the held-out test split (4,641 pairs, 1,470 positive). The two DeepImmuno rows are different models: the released checkpoint, and the same architecture retrained on our training split. Within-allele AUC is computed inside each allele, so the allele prior scores 0.500 by construction.

Predictor	Pooled AUC	Within-allele AUC
This study	0.832	0.769
Gradient boosting on engineered features	0.810	0.721
DeepImmuno architecture, retrained here	0.792	0.661
Allele prior (no peptide information)	**0.752**	0.500
BigMHC_IM (Albert et al., 2023)	0.647	0.609
DeepImmuno (Li et al., 2021)	0.606	0.633
MixMHCpred 3.0 (binding only) (Tadros et al., 2025)	0.559	0.570

3.6 Published predictors were trained on the evaluation data

Intersecting our evaluation pairs with released training sets (§7.5) shows substantial overlap, and performance falls sharply outside it (**Table 6**).

Table 6. Overlap between our evaluation pairs and the released training sets of two published predictors, with performance inside and outside the overlap. Overlap is counted by shared peptide. MixMHCpred's training set is not public and could not be audited.

Predictor	Evaluation pairs it had seen	Seen	Unseen
DeepImmuno (Li et al., 2021)	37.5% (shared peptide)	0.746	**0.512**
BigMHC_IM (Albert et al., 2023)	27.1% (shared peptide)	0.807	**0.549**

On the 778 lung-cancer-associated pairs whose sequence appears in neither training set, every published predictor's interval includes chance: DeepImmuno 0.528 (0.476–0.583), BigMHC 0.502 (0.438–0.561), MixMHCpred 0.446 (0.383–0.502), allele prior 0.465 (0.415–0.514).

3.7 The definition of a negative changes the task by ≈0.4 AUC

Using the per-patient supplementary table of the harmonised multi-cancer resource of Müller et al. (2023), which covers 99 patients (the original report also includes an in-house cohort, for 131 in all), the negative-set definition alone moves per-patient AUC by about 0.4 (**Table 7**).

Table 7. Effect of the negative-set definition on per-patient AUC. The same 178 immunogenic peptides are ranked in both rows; only the set they are ranked against differs.

What counts as a negative	Peptides compared	Per-patient AUC
Every candidate neo-peptide, screened or not	1,780,481 (76% never assayed)	≈ 0.97
Assayed negatives only	**421,017 assayed (23.6% of candidates)**	**≈ 0.55**

Of the 421,017 screened peptides, 178 were immunogenic. Ranking those 178 against *all* candidates gives per-patient AUC ≈ 0.97; discriminating them from assayed negatives gives ≈ 0.55. Both are published as immunogenicity prediction. Most "negatives" in the former are peptides nobody screened, overwhelmingly non-binders that any filter rejects.

3.8 Leakage-free performance approaches the measured ceiling, and architecture makes no measurable difference

We scored every model from study-grouped out-of-fold predictions, within study and allele, across 205 groups, bootstrapping over studies (**Figure 2**, panel b; **Table 8**).

All intervals clear chance, and the best result is 84% of the conservative reliability bound (0.681). Given identical pretraining, folds and seeds, the convolutional model differs from the attention model by +0.013 (95% CI −0.017 to +0.043), which is not significant. Architecture made no measurable difference. Only one comparison reaches significance: the ranking objective beats pointwise training by +0.035 (CI +0.001 to +0.069). The lower bound sits almost on zero, so we call that result marginal.

Study-aware reweighting recovers much of the loss on pooled metrics (0.623 → 0.700) and costs less than a bespoke objective.

Table 8. Within-study and within-allele AUC for six training schemes, scored from study-grouped out-of-fold predictions across 205 groups. Intervals bootstrap over studies. All schemes share pretraining, folds, class weighting and seeds, so only the objective differs.

Training scheme	AUC	95% CI
Ranking objective, no engineered features	**0.573**	0.544–0.603
Ranking objective, with features	0.557	0.526–0.588
CNN (original Conv2D architecture)	**0.544**	0.516–0.572
Study reweighting	0.548	0.519–0.576
Pointwise, no features	0.538	0.511–0.566
Pointwise baseline	0.532	0.504–0.560

3.9 A worked example: a feature that standard validation endorses and grouped validation rejects

Self-similarity to the human proteome looks strongly predictive under standard validation. Pooled across lung-cancer-associated pairs it reaches AUC 0.671 (CI 0.630–0.712), and under peptide-grouped validation it raised within-allele AUC by +0.034.

Grouping by study removes the effect. Within study its AUC is 0.483, and no study scores above chance. Withholding the whole feature block changes within-study AUC by +0.005 (CI −0.015 to +0.024) and within-study-and-allele AUC by −0.007; neither is significant. The feature tells studies apart, and says nothing about which peptides within a study are immunogenic.

Of nine feature families tested within study, only two clear chance: mean hydrophobicity 0.550 (0.518–0.582) and aromatic content 0.542 (0.517–0.568), both consistent with known determinants of TCR recognition (Calis et al., 2013).

4. Discussion

Reported AUCs of 0.85 and above (**Table 1**) are higher than the measured reliability of the labels (0.636, or 0.681 under the conservative bound) and higher than any model we could build under leakage-free evaluation (0.573). We did not reproduce the published figures on their own test sets, so what follows compares evaluation schemes on our data. We attribute the gap to three things: peptide-level splitting, which lowers AUC by only 0.014 relative to random splitting; training-set overlap, which we found in both audited predictors; and negatives defined as untested candidates, which changes the task.

AUC is measured against noisy labels. A model that learns only transferable biology should therefore score near the label reliability, and one that scores well above it is probably also fitting the noise. Our results place that noise at the level of the study.

Alternative explanations. Four readings of §3.4 would weaken the argument, and the checks there address each. If the peptide split were merely too lenient about similarity, a homology-aware split would recover the study-grouped number; it does not, since sequence clusters move AUC by under 0.02. If the effect were specific to gradient boosting, logistic regression would differ; it follows the same ordering. If assay differences drove it, it would vanish within an assay family; it does not. If one study drove it, removing that study would remove the effect; it removes about 60% of the pooled gap and leaves 0.09. A forward-in-time test, the nearest this benchmark comes to deployment, gives 0.563 for gradient boosting, close to the leakage-free estimate, though the interval (0.442–0.684) is wide because half of the test pairs come from ten studies.

Agreement as a bound. Agreement between annotators does not cap model performance in general. If annotators err independently, a well-specified model can beat their mutual agreement (Richie et al., 2022). Our figures therefore describe performance against these labels, not how accurate prediction could be in principle. Random label noise would make a high measured AUC unattainable. High AUCs are attainable here, which indicates that the noise is systematic and learnable, as §3.4–3.6 show.

Relation to prior work. Kapoor and Narayanan (2023) documented this pattern across 17 fields, and ours is an instance in computational immunology. Whalen et al. (2022) catalogue the same failure modes in genomics, naming distributional differences between training and test data and unaccounted-for confounders as the two that most often inflate reported performance; source publication is both at once here. In their case study, complex models claimed to beat logistic regression and ceased to do so once leakage was corrected; in ours, published predictors are outscored by a baseline that cannot see the peptide. Our setting also lets us measure the reliability of the labels directly instead of assuming it, because IEDB records which publication produced each measurement and some pairs were measured more than once. We extend Lee et al. (2023), who identified HLA bias and cross-species degradation, by identifying the source publication as the operative grouping unit and by

quantifying the label ceiling. Recent tools have begun to hold out homologous sequences (Kim et al., 2025). Our results suggest that this is not sufficient: sequence clusters moved AUC by under 0.02, whereas holding out publications moved it by more than 0.2.

What bounds performance here. At 0.573 against a conservative bound of 0.681, we are at 84% of what the labels permit, leaving ≈0.11 AUC of headroom. We do not think architecture can recover it: the convolutional and attention models do not differ significantly, and neoantigen methods that rank better draw on information that sequence-only benchmarks lack, such as HLA presentation hotspots, binding promiscuity and the oncogenic role of the mutated gene (Müller et al., 2023). Methods that add TCR sequence without patient context still struggle to generalize to unseen epitopes (Lu et al., 2026), so the limit is not specific to peptide-only models.

Raising the ceiling requires harmonised assays of the kind whose adoption improved performance between two successive proficiency panels (Janetzki et al., 2008). Broadening coverage requires more distinct epitopes, currently limited by peptide-library production, and patient-specific prediction requires patient-specific measurement.

4.1 Recommendations

These extend, for this task, the general reporting items of the DOME recommendations for supervised learning in biology (Walsh et al., 2021), which ask for the data partitioning scheme to be stated but do not prescribe a grouping unit.

1. Split by source publication, not only by peptide, and state the grouping unit used. Grouping by sequence similarity is not a substitute.
2. Report within-study and within-allele AUC alongside any pooled figure.
3. Include a peptide-blind allele-prior baseline. If it is competitive, the benchmark is measuring dataset structure.
4. State the negative-set definition explicitly; untested candidates and assayed negatives are different tasks.
5. Audit training-set overlap before reporting comparative performance.
6. Bootstrap over studies, not rows; peptides within a study are not independent.
7. Report whether the result holds within an assay family and without the largest single study.

5. Limitations

- Label reliability is measured on IEDB alone. The independent dataset available to us lacks publication identifiers, so §3.4 but not §3.2 could be replicated externally.
- Repeat-tested pairs are a selected subset. We address this by stratification and adopt the most favourable bound, but the estimate remains conditional on that subset.
- We did not reproduce the published values of Table 1 on the datasets and test sets of the papers that report them. The comparison is between evaluation schemes on our data, and between predictors on our pairs, not an audit of specific published numbers.
- Study-grouped evaluation could not be replicated on an independent dataset. No public immunogenicity dataset we found records the source publication of each label, and TRAP supports antigen and organism grouping only.
- About 60% of the pooled random-to-study gap depends on one all-negative study (§3.4). The gap would differ in a benchmark with another composition.
- The forward-in-time check covers 58% of pairs, those with a PubMed ID, uses ID order as a proxy for publication date, and its test pairs are concentrated in a few studies (50% come from ten), so its intervals are wide.
- MixMHCpred's training set is not public and could not be audited for contamination.
- Our protocol is stricter than the usual one, but we do not show that it predicts clinical success better. That would require prospective experimental validation.
- Analysis is restricted to 9–10mers and HLA-A/B/C, inherited from the encoding of the predictor we reimplemented.
- We propose no improved predictor. Our best model is a leakage-free reference point, and its apparent advantage over published tools on the peptide-grouped split (0.832 against 0.606 for DeepImmuno's released checkpoint) is an artifact of the split we argue against, and of our model having been fitted to this dataset's study structure.

6. Conclusion

The labels underlying T-cell epitope immunogenicity benchmarks reproduce across laboratories at AUC 0.636 (κ = 0.273; conservatively 0.681). Published predictors report 0.85 and above under random cross-validation, and 0.56 to 0.68 on independently constructed test sets (**Table 1**). We find three features of benchmark construction that inflate apparent performance: peptide-level splitting that removes almost none of the leakage (a 0.014 drop in AUC against random splitting), training-set contamination of 27–38%, and negative-set definitions that alter the task being measured. Most of the leakage is study structure rather than sequence similarity, and about 60% of its size in this benchmark comes from one all-negative study. A leakage-free model reaches 0.573, attaining 84% of what the labels permit, and, trained identically, the convolutional model scores 0.013 above the attention model (95% CI -0.017 to +0.043), which is not significant. Progress in this area is currently limited by measurement reproducibility and epitope coverage, and model architecture is not the constraint.

7. Methods

7.1 Data retrieval and cleaning

T-cell assay records were retrieved from the IEDB (Vita et al., 2019) Query API restricted to MHC class I, human host, and linear peptides of length 9–10 (92,721 records). Records were required to carry a four-digit HLA-A/B/C allele present in our pseudo-sequence table and a canonical twenty-letter peptide, leaving 65,732 records over 32,167 unique pairs. Any *Positive** qualitative measure was mapped to 1 and *Negative* to 0. Records were collapsed to one label per pair by majority vote; 893 exact ties were dropped, giving 31,274 pairs, 31.7% positive. A further 400,000 MHC binding and ligand-elution records were retrieved under the same constraints for encoder pretraining.

7.2 Split schemes

Two schemes are stored in the same table: *split*/*fold* grouped by peptide, and *study_split*/*study_fold* grouped by *reference_id*. Each uses a 15% grouped hold-out and 10 grouped CV folds.

The checks in §3.4 add three similarity-based grouping rules, each used with five grouped folds. Peptides of equal length within one or two mismatches are merged by single linkage. Candidate pairs come from a pigeonhole index (two peptides within d mismatches must agree exactly on one of d+1 segments) and are then checked exactly. Peptides sharing any exact six-residue substring are also merged; a five-residue rule was rejected because it joined 14.5% of all pairs into one cluster. Peptides of different lengths are never merged. For the forward-in-time test each study is dated by its smallest PubMed ID, and the training set is every pair from studies up to the 80th percentile of dated pairs. A pair counts as belonging to an assay family only if all of its records come from that family, the family being the last item of the IEDB assay name.

7.3 Reliability measurement

Pairs assayed in ≥2 distinct publications were identified. Each study contributes its majority outcome for that pair; within-study ties are dropped. Two studies are drawn at random per pair, and we compute agreement, Cohen's κ, and the AUC of study A's label predicting study B's. The point estimate is the mean over 200 random pairings of the full set, so it does not depend on one arbitrary draw. The interval is a separate bootstrap that resamples the pairs themselves with replacement, 2,000 times, drawing a fresh pairing each time; percentiles taken across re-pairings of one fixed set of pairs would capture only how the two studies are chosen and would be about half as wide. Because both labels are binary, this AUC equals balanced accuracy. Sensitivity was assessed over four aggregation rules plus an unaggregated variant, over assay families, over HLA locus, and stratified by the number of studies testing a pair.

7.4 Models

Published predictors. DeepImmuno (Li et al., 2021) (two-branch Conv2D), scored through its released *cnn_model_331_3_7* checkpoint, which we verified byte-identical to the copy in this repository; BigMHC_IM (Albert et al., 2023) (seven-network ensemble, presentation-pretrained then transfer-learned); MixMHCpred 3.0 (Tadros et al., 2025), binding only, included as a control. PRIME 2.0 (Gfeller et al., 2023) was obtained but its bundled Windows binary did not execute; it is excluded and documented. Where a table reports the DeepImmuno architecture retrained by us, it is labelled as such and is a different set of weights from the released checkpoint.

Baselines. An allele prior predicting each allele's training-set positive rate, using no peptide information; gradient boosting on the AAindex/PCA encoding.

This study. A bidirectional cross-attention network (65,987 parameters): learned residue and position embeddings, self-attention within the 10-position peptide and the 46-residue HLA pseudo-sequence, then cross-attention in both directions, a dense trunk and a sigmoid output. Pretrained on the binding corpus (held-out binding AUC 0.958) and fine-tuned on immunogenicity.

Training schemes. Pointwise cross-entropy; dropping the dominant study; study-size reweighting; training on mixed-outcome studies only; and a study-stratified pairwise ranking objective over (immunogenic, non-immunogenic) pairs sharing a study and an allele. The CNN was trained from fresh initialisation, never from the published checkpoint, with identical pretraining, folds, class weighting and seeds.

7.5 Evaluation

We report ROC-AUC throughout, for comparability with the figures of **Table 1**. ROC-AUC and PR-AUC can rank methods differently when classes are imbalanced (Saito and Rehmsmeier, 2015); PR-AUC for every predictor is included in the released result files rather than tabulated here. Pooled AUC is reported with two stricter quantities: within-allele AUC, computed inside each allele; and within-study and within-study-and-allele AUC, computed inside each source publication. Groups require ≥10 pairs and both outcome classes. Confidence intervals for within-study quantities bootstrap over studies rather than rows. Models are scored from study-grouped out-of-fold predictions. Differences between grouping rules, and the forward-in-time intervals, resample whole studies with replacement (1,000 replicates), because pairs within a study are not independent. Differences are computed from unrounded values, so they can differ by 0.001 from the difference of two rounded figures in a table. Published training sets were intersected with our evaluation pairs by exact peptide and by exact peptide–HLA pair.

Data and code availability

All analyses are reproducible from *docs/PIPELINE.md*, and *notebooks/Reproduce_Manuscript.ipynb* reproduces every number and figure. Code is organised as *src/data* (retrieval, cleaning, splits), *src/models* (architectures, training) and *src/analysis* (all reported analyses). The curated benchmark, both split schemes and all model predictions are released with the repository, together with the outputs of every audited third-party predictor on our pairs, so the comparisons in §3.5 and §3.6 can be reproduced without installing those tools. DeepImmuno's released checkpoint is scored directly; *python -m src.models.deepimmuno* prints its held-out AUC in one command.

Archived at Zenodo: *10.5281/zenodo.23719454* for all versions, *10.5281/zenodo.23719455* for v1.0.0, the release that produced every number reported here, and *10.5281/zenodo.23840985* for v1.0.1, which adds the consistency checks and provenance files behind this manuscript without changing any reported number. Repository: *https://github.com/Terano24/Deep-Learning-Research-Project*. Code is released under the MIT licence; redistributed third-party data retains its own terms, listed in *LICENSE*.

External resources: IEDB Query API; UniProt UP000005640; TRAP (Lee et al., 2023); the harmonised datasets of Müller et al. (2023); the public releases of DeepImmuno (Li et al., 2021), BigMHC (Albert et al., 2023) and MixMHCpred (Tadros et al., 2025).

CRediT authorship contribution statement

I Nyoman Agung Ananda Terano: Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Mirrabell Frederica Hadiwijono:** Conceptualization, Methodology, Investigation, Validation, Writing – original draft, Writing – review & editing.

Both authors contributed to study design and manuscript preparation, read and approved the final manuscript, and accept responsibility for its content. I Nyoman Agung Ananda Terano is the corresponding author and handles submission and correspondence.

Competing interests

The authors declare no competing interests.

Funding

This research received no specific grant from any funding agency.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to draft and edit the manuscript text. The research question, the study design, the choice of analyses and the interpretation of the results are the authors' own. The authors reviewed and edited all AI-assisted content and take full responsibility for the content of the published article.

References

Albert, B.A., Yang, Y., Shao, X.M., Singh, D., Smith, K.N., Anagnostou, V., Karchin, R., 2023. Deep neural networks predict class I major histocompatibility complex epitope presentation and transfer learn neoepitope immunogenicity. Nat. Mach. Intell. 5 (8), 861–872. https://doi.org/10.1038/s42256-023-00694-6

Britten, C.M., Janetzki, S., Ben-Porat, L., Clay, T.M., Kalos, M., Maecker, H., Odunsi, K., Pride, M., Old, L., Hoos, A., Romero, P., 2009. Harmonization guidelines for HLA-peptide multimer assays derived from results of a large scale international proficiency panel of the Cancer Vaccine Consortium. Cancer Immunol. Immunother. 58 (10), 1701–1713. https://doi.org/10.1007/s00262-009-0681-z

Calis, J.J.A., Maybeno, M., Greenbaum, J.A., Weiskopf, D., De Silva, A.D., Sette, A., Peters, B., Keşmir, C., 2013. Properties of MHC class I presented peptides that enhance immunogenicity. PLoS Comput. Biol. 9 (10), e1003266. https://doi.org/10.1371/journal.pcbi.1003266

Cohen, J., 1960. A coefficient of agreement for nominal scales. Educ. Psychol. Meas. 20 (1), 37–46. https://doi.org/10.1177/001316446002000104

Farriol-Duran, R., Domínguez-Dalmases, C., Cañellas-Solé, A., Vazquez, M., Porta-Pardo, E., Guallar, V., 2025. PredIG: an interpretable predictor of T-cell epitope immunogenicity. Genome Med. 17, 140. https://doi.org/10.1186/s13073-025-01569-8

Gfeller, D., Schmidt, J., Croce, G., Guillaume, P., Bobisse, S., Genolet, R., Queiroz, L., Cesbron, J., Racle, J., Harari, A., 2023. Improved predictions of antigen presentation and TCR recognition with MixMHCpred2.2 and PRIME2.0 reveal potent SARS-CoV-2 $CD8^+$ T-cell epitopes. Cell Syst. 14 (1), 72–83.e5. https://doi.org/10.1016/j.cels.2022.12.002

Janetzki, S., Panageas, K.S., Ben-Porat, L., Boyer, J., Britten, C.M., Clay, T.M., Kalos, M., Maecker, H.T., Romero, P., Yuan, J., Kast, W.M., Hoos, A., 2008. Results and harmonization guidelines from two large-scale international Elispot proficiency panels conducted by the Cancer Vaccine Consortium (CVC/SVI). Cancer Immunol. Immunother. 57 (3), 303–315. https://doi.org/10.1007/s00262-007-0380-6

Jiang, D., Qi, Y., Zhang, H., Zhang, Y., Zhang, L., Wang, J., Chen, J., Zhu, W., Li, Y., Liu, J., 2024. NeoaPred: a deep-learning framework for predicting immunogenic neoantigen based on surface and structural features of peptide–human leukocyte antigen complexes. Bioinformatics 40 (9), btae547. https://doi.org/10.1093/bioinformatics/btae547

Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns 4 (9), 100804. https://doi.org/10.1016/j.patter.2023.100804

Kim, J., Jung, N., Lee, J., Cho, N.H., Noh, J., Seok, C., 2025. T-SCAPE: T cell immunogenicity scoring via cross-domain aided predictive engine. Sci. Adv. 11 (49). https://doi.org/10.1126/sciadv.adz8759

Landis, J.R., Koch, G.G., 1977. The measurement of observer agreement for categorical data. Biometrics 33 (1), 159–174. https://doi.org/10.2307/2529310

Lee, C.H., Huh, J., Buckley, P.R., Jang, M., Pinho, M.P., Fernandes, R.A., Antanaviciute, A., Simmons, A., Koohy, H., 2023. A robust deep learning workflow to predict $CD8^+$ T-cell epitopes. Genome Med. 15, 70. https://doi.org/10.1186/s13073-023-01225-z

Li, G., Iyer, B., Prasath, V.B.S., Ni, Y., Salomonis, N., 2021. DeepImmuno: deep learning-empowered prediction and generation of immunogenic peptides for T-cell immunity. Brief. Bioinform. 22 (6), bbab160. https://doi.org/10.1093/bib/bbab160

Lu, Y., Wang, Y., Xu, M., Xie, B., Yang, Y., Xu, H., Suo, S., 2026. Assessment of computational methods in predicting TCR–epitope binding recognition. Nat. Methods 23 (1), 248–259. https://doi.org/10.1038/s41592-025-02910-0

Müller, M., Huber, F., Arnaud, M., Kraemer, A.I., Ricart Altimiras, E., Michaux, J., Taillandier- Coindard, M., Chiffelle, J., Murgues, B., Gehret, T., Auger, A., Stevenson, B.J., Coukos, G., Harari, A., Bassani-Sternberg, M., 2023. Machine learning

methods and harmonized datasets improve immunogenic neoantigen prediction. Immunity 56 (11), 2650–2663.e6. https://doi.org/10.1016/j.immuni.2023.09.002

O'Brien, H., Salm, M., Morton, L.T., Szukszto, M., O'Farrell, F., Boulton, C., King, L., Bhuva, M., Reading, J.L., Litchfield, K., 2024. A modular protein language modelling approach to immunogenicity prediction. PLoS Comput. Biol. 20 (11), e1012511. https://doi.org/10.1371/journal.pcbi.1012511

O'Donnell, T.J., Rubinsteyn, A., Laserson, U., 2020. MHCflurry 2.0: improved pan-allele prediction of MHC class I-presented peptides by incorporating antigen processing. Cell Syst. 11 (1), 42–48.e7. https://doi.org/10.1016/j.cels.2020.06.010

Ott, P.A., Hu, Z., Keskin, D.B., Shukla, S.A., Sun, J., Bozym, D.J., Zhang, W., Luoma, A., Giobbie-Hurder, A., Peter, L., et al., 2017. An immunogenic personal neoantigen vaccine for patients with melanoma. Nature 547 (7662), 217–221. https://doi.org/10.1038/nature22991

Reynisson, B., Alvarez, B., Paul, S., Peters, B., Nielsen, M., 2020. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. Nucleic Acids Res. 48 (W1), W449–W454. https://doi.org/10.1093/nar/gkaa379

Richie, R., Grover, S., Tsui, F.R., 2022. Inter-annotator agreement is not the ceiling of machine learning performance: evidence from a comprehensive set of simulations. In: Proceedings of the 21st Workshop on Biomedical Language Processing (BioNLP), ACL, pp. 275–284. https://doi.org/10.18653/v1/2022.bionlp-1.26

Rojas, L.A., Sethna, Z., Soares, K.C., Olcese, C., Pang, N., Patterson, E., Lihm, J., Ceglia, N., Guasp, P., Chu, A., et al., 2023. Personalized RNA neoantigen vaccines stimulate T cells in pancreatic cancer. Nature 618 (7963), 144–150. https://doi.org/10.1038/s41586-023-06063-y

Sahin, U., Derhovanessian, E., Miller, M., Kloke, B.P., Simon, P., Löwer, M., Bukur, V., Tadmor, A.D., Luxemburger, U., Schrörs, B., et al., 2017. Personalized RNA mutanome vaccines mobilize poly-specific therapeutic immunity against cancer. Nature 547 (7662), 222–226. https://doi.org/10.1038/nature23003

Saito, T., Rehmsmeier, M., 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS One 10 (3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Tadros, D.M., Racle, J., Gfeller, D., 2025. Predicting MHC-I ligands across alleles and species: how far can we go? Genome Med. 17 (1). https://doi.org/10.1186/s13073-025-01450-8

Vita, R., Mahajan, S., Overton, J.A., Dhanda, S.K., Martini, S., Cantrell, J.R., Wheeler, D.K., Sette, A., Peters, B., 2019. The Immune Epitope Database (IEDB): 2018 update. Nucleic Acids Res. 47 (D1), D339–D343. https://doi.org/10.1093/nar/gky1006

Walsh, I., Fishman, D., Garcia-Gasulla, D., Titma, T., Pollastri, G., ELIXIR Machine Learning Focus Group, et al., 2021. DOME: recommendations for supervised machine learning validation in biology. Nat. Methods 18 (10), 1122–1127. https://doi.org/10.1038/s41592-021-01205-4

Wang, G., Wan, H., Jian, X., Li, Y., Ouyang, J., Tan, X., Zhao, Y., Lin, Y., Xie, L., 2020. INeo-Epp: a novel T-cell HLA class-I immunogenicity or neoantigenic epitope prediction method based on sequence-related amino acid features. BioMed Res. Int. 2020, 5798356. https://doi.org/10.1155/2020/5798356

Weber, J.S., Carlino, M.S., Khattak, A., Meniawy, T., Ansstas, G., Taylor, M.H., Kim, K.B., McKean, M., Long, G.V., Sullivan, R.J., et al., 2024. Individualised neoantigen therapy mRNA-4157 (V940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (KEYNOTE-942): a randomised, phase 2b study. Lancet 403 (10427), 632–644. https://doi.org/10.1016/S0140-6736(23)02268-7

Wells, D.K., van Buuren, M.M., Dang, K.K., Hubbard-Lucey, V.M., Sheehan, K.C.F., Campbell, K.M., Lamb, A., Ward, J.P., Sidney, J., Blazquez, A.B., et al., 2020. Key parameters of tumor epitope immunogenicity revealed through a consortium approach improve neoantigen prediction. Cell 183 (3), 818–834.e13. https://doi.org/10.1016/j.cell.2020.09.015

Whalen, S., Schreiber, J., Noble, W.S., Pollard, K.S., 2022. Navigating the pitfalls of applying machine learning in genomics. Nat. Rev. Genet. 23 (3), 169–181. https://doi.org/10.1038/s41576-021-00434-9